

1st REVIEW

Online Food Delivery Platform: Multi-Restaurant Ordering & Unified Checkout

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ABSTRACT

- Online food delivery platforms have become an essential part of modern food ordering, but most existing systems require customers to place separate orders when purchasing items from different restaurants. This results in repeated checkout processes, separate delivery charges, multiple order statuses, and a fragmented tracking experience. The proposed Online Food Delivery Platform addresses this limitation by enabling customers to browse multiple restaurants, select items from different restaurants, add them to a single cart, and complete one unified checkout and payment process. The system is designed to provide a convenient and transparent ordering experience while reducing the complexity associated with managing multiple restaurant orders.
- Behind the unified customer experience, the platform manages the order through a master-order mechanism that automatically divides the order into independent restaurant-specific orders. Each restaurant can access and process only its respective portion of the order, while delivery partners are assigned to fulfill the deliveries and update their status. The platform provides role-based dashboards for customers, restaurants, delivery partners, and administrators, along with real-time order and delivery status updates. The system is developed using React.js for the frontend, Node.js and Express.js for the backend, MySQL for relational data management, and Socket.io for real-time communication, with Google Maps API and x402 Aragen supporting delivery tracking and online payments respectively.

INTRODUCTION

- Online food delivery has become a widely used service, providing customers with convenient access to food from restaurants through digital platforms. However, when customers want to order items from multiple restaurants, most existing platforms require them to place separate orders for each restaurant. This leads to multiple checkouts, separate delivery charges, different order IDs, and difficulty in monitoring the status of all orders simultaneously. These limitations create an opportunity for a more integrated food ordering experience that can combine orders from multiple restaurants while maintaining efficient restaurant-wise processing. The platform serves four types of users through role-based dashboards:
 - Customers — browse restaurants, build a multi-restaurant cart, and check out once
 - Restaurants — manage their menu and process only their own portion of an order
 - Delivery Partners — accept and fulfill deliveries with live location tracking
 - Admin — oversees users, restaurants, delivery partners, commissions, and offers

EXISTING SYSTEM

Most existing online food delivery platforms follow a **single-restaurant ordering model**, where a customer's cart is associated with one restaurant at a time. To order from another restaurant, the customer must create a separate order and complete another checkout process.

- **Drawbacks / Limitations:**
- **Higher Food Prices** — Menu prices on delivery platforms may be higher than the prices offered directly by restaurants.
- **Multiple Additional Charges** — Platform, handling, packaging, convenience, and other charges can increase the final bill.
- **High Delivery Charges** — Delivery fees may vary based on distance, demand, peak hours, and other factors.
- **Unclear Final Pricing** — The final payable amount may increase considerably after additional charges and taxes are applied.
- **Variable Delivery Times** — Estimated delivery times may change due to restaurant preparation, traffic, and delivery conditions.
- **Limited Ordering Flexibility** — The ordering process is primarily designed around individual restaurant orders.

PROBLEM

- Customers who want food items from **multiple restaurants** cannot conveniently combine them into a single order. They are required to place separate orders, resulting in **multiple checkouts, separate delivery charges, multiple order statuses, and fragmented tracking**.
- This makes the overall ordering process more expensive, time-consuming, and difficult to manage when customers want items from different restaurants.
- **Goal:**
To provide a **unified multi-restaurant ordering experience** with one cart, one checkout, consolidated payment, and a single view for tracking the entire order.

SOLUTION(Contribution)

- The proposed **Online Food Delivery Platform** introduces a multi-restaurant ordering model that allows customers to combine items from different restaurants into a single order.

Key Contributions:

- **One Unified Cart** — Customers can add items from multiple restaurants to one cart.
- **Single Checkout & Payment** — A single checkout process is used for the combined order.
- **Master Order Management** — The system creates one master order and automatically splits it into restaurant-specific orders.
- **Independent Restaurant Processing** — Each restaurant receives and processes only its respective portion of the order.
- **Unified Order Tracking** — Customers can view the status of their complete order from one place.
- **Delivery Partner Assignment** — Deliveries are assigned and managed through the platform.
- **Real-Time Updates** — Order and delivery status are updated in real time.

Objectives

- To develop a platform that allows customers to **browse and order from multiple restaurants in a single session.**
- To provide a **unified cart and single checkout process** for items selected from different restaurants.
- To automatically **split a master order into restaurant-specific orders** for independent processing and fulfillment.
- To provide restaurants with a dedicated dashboard to **manage menus, receive orders, and update order status.**
- To enable delivery partners to **accept assigned deliveries and provide real-time delivery status updates.**
- To provide customers with **consolidated order tracking** across all restaurant-specific orders.
- To provide administrators with a centralized system to **manage users, restaurants, delivery partners, orders, commissions, and offers.**

Modules

- **1. Customer Module**

- Browse and search restaurants
- Add items from multiple restaurants to cart
- Unified checkout and payment
- Track complete order status
- Ratings and reviews

- **2. Restaurant Module**

- Manage restaurant menu
- Receive restaurant-specific orders
- Accept and process orders
- Update order status
- Manage offers and discounts

- **3. Delivery Partner Module**

- View assigned deliveries
- Accept delivery requests
- Manage pickup and delivery
- Update delivery status
- Live location tracking

- **4. Admin Module**

- Manage customers, restaurants, and delivery partners
- Monitor orders
- Manage commissions
- Manage offers and discounts

SOFTWARE REQUIREMENTS

- **Frontend**

- React.js — UI for customer, restaurant, delivery, and admin dashboards

- **Backend**

- Node.js — JavaScript runtime powering the server
- Express.js — REST API framework for all backend routes

- **Database**

- MySQL — relational database for orders, users, and restaurants

- **Real-Time & Integrations**

- Socket.io — real-time order status and live updates
- Google Maps API — location services and delivery tracking
- x402 Aragen — secure online payment gateway

- System Architecture: Users (Customer / Restaurant / Delivery / Admin) → React.js → Node.js + Express.js → MySQL ⇌ Socket.io

CONCLUSION AND FUTURE SCOPE

- The platform replaces multiple, disconnected single-restaurant orders with one unified cart, one checkout, and one order experience spanning several restaurants.
- A master order is transparently split into restaurant-specific orders, so each kitchen fulfills only its own portion while the customer tracks everything in one place.
- Built on React.js, Node.js, Express.js, MySQL, and Socket.io, the system delivers real-time order status, live delivery tracking, and role-based dashboards for customers, restaurants, delivery partners, and admin.
- **Future Scope**
 - Scale the platform to support more restaurants, delivery partners, and cities
 - Introduce AI-based restaurant and dish recommendations
 - Add loyalty/rewards programs and additional payment options
 - Extend analytics for admin around order patterns and delivery performance

THANK YOU

Questions & Discussion